

What is a Magnetic Field?

A magnetic field is a region where magnetic forces act on certain materials.

It is generated by:

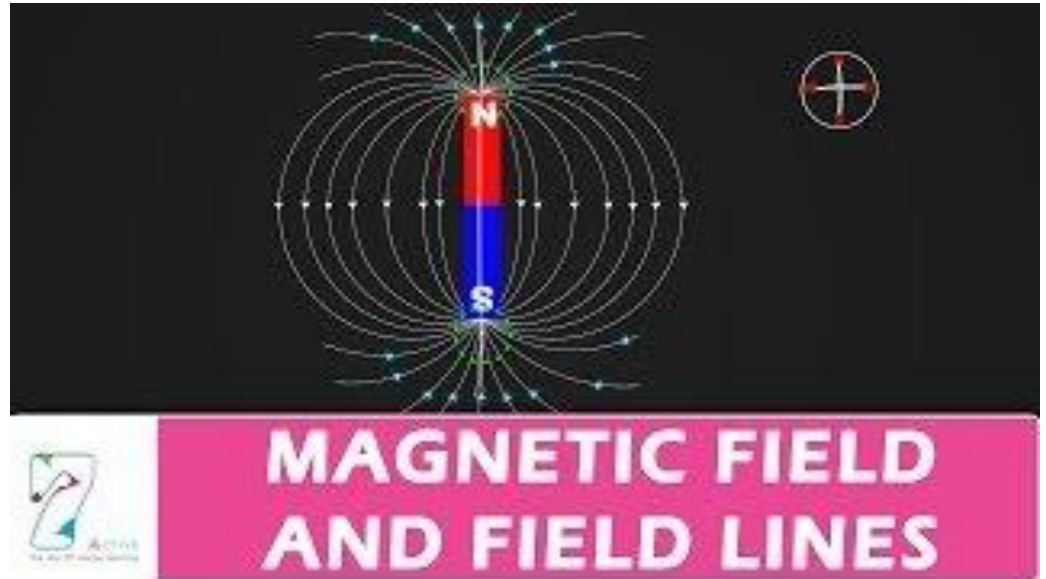
- Moving electric charges (current in a wire).
- Permanent magnets (due to atomic magnetic moments).

Magnetic field lines show:

- The direction of the magnetic force.
- The relative strength of the field (closer lines = stronger field).

Magnetic fields can influence materials that have magnetic properties, such as iron filings.

The force acts on magnetizable materials, not on all materials.



Magnetic fields are invisible, but their effect is visible when they align small magnetic dipoles like those in iron filings.

Magnetization: Induced and Remanent

Materials can acquire two types of magnetization:

- Induced magnetization: Temporary, only exists while an external field is present.
- Remanent magnetization: Permanent, remains after the external field is removed.

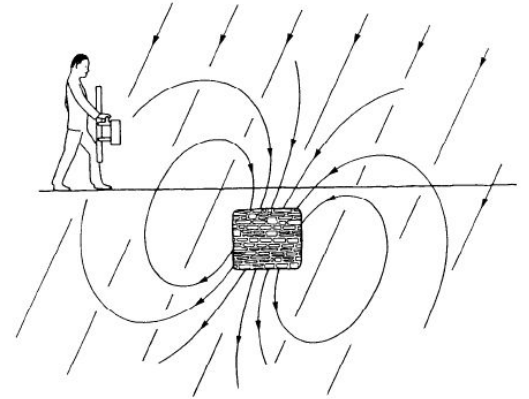
Induced magnetization is the foundation of magnetic prospecting.

The induced magnetization depends on:

- The strength of the Earth's magnetic field.
- The magnetic susceptibility of the material.

Magnetic susceptibility describes how easily a material can be magnetized when exposed to an external magnetic field.

$$M = \chi H \quad B = \mu_0(H + M)$$



In magnetic prospecting, we mainly measure the induced magnetization created by the Earth's field in subsurface materials

Magnetic Susceptibility: What Causes It?

The magnetic properties of materials are determined at the atomic level.

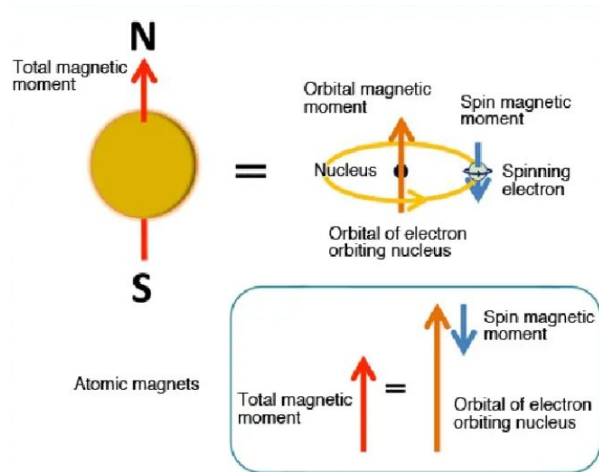
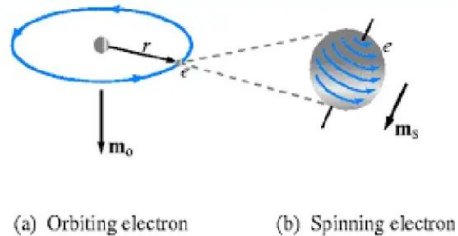
Magnetic susceptibility arises from the magnetic moments of atoms.

Magnetic moments are caused by:

- The orbital motion of electrons around the nucleus.
- The spin of electrons.
- The number of unpaired electrons in electron shells.

These effects are inherently quantum mechanical.

As a result, each atom behaves like a small magnetic dipole.



Magnetic Susceptibility: What Causes It?

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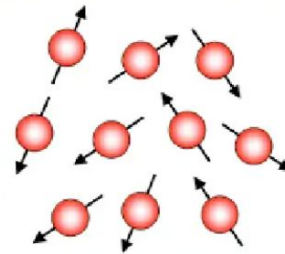
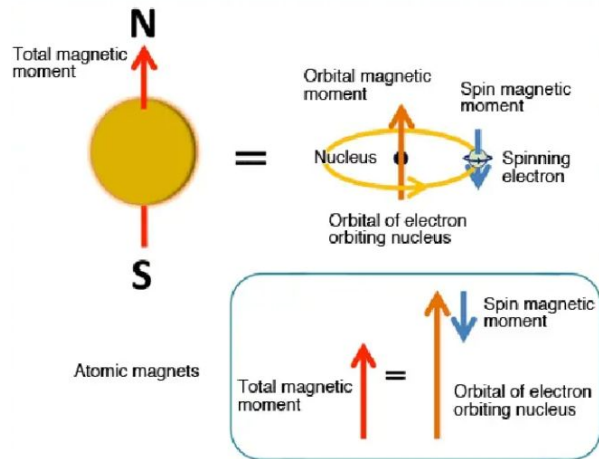
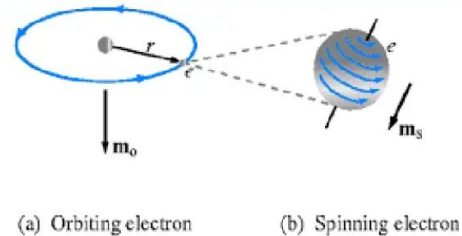
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The alignment and interaction of these atomic dipoles define whether the material is diamagnetic, paramagnetic, or ferromagnetic.

Magnetic Fundamentals

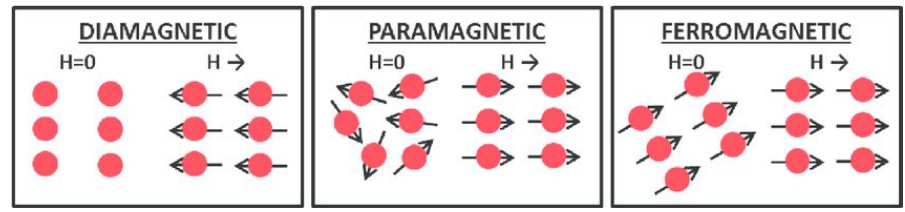
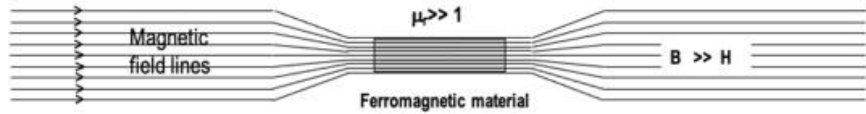
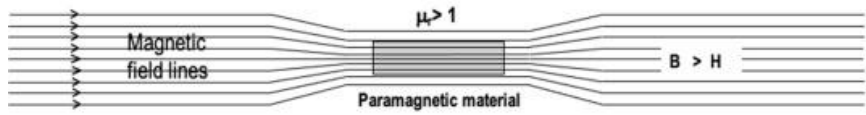
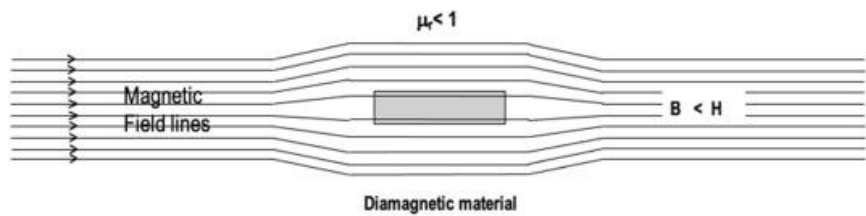
Types of Magnetic Materials

Materials are classified based on their response to the applied field:

- **Diamagnetic:**
Negative susceptibility.
Diamagnetic materials align their induced field opposite to the applied field.
Weakly repelled by magnetic fields.
Example: Quartz.
- **Paramagnetic:**
Positive, weak susceptibility.
Paramagnetic materials align their induced field in the same direction as the applied field.
Slightly attracted to magnetic fields.
Example: Olivine, Biotite.
- **Ferromagnetic:**
Strong positive susceptibility.
Ferromagnetic materials exhibit strong alignment with the external field
can remain magnetized even after the external field is removed.
Example: Magnetite, Iron.

Magnetic susceptibility describes how easily a material can be magnetized when exposed to an external magnetic field.

$$M = \chi H \qquad B = \mu_0(H + M)$$



Magnetic Susceptibility of Common Materials

Although the mechanisms by which induced magnetization can arise are rather complex, the field generated by these mechanisms can be quantified by a single, simple parameter known as the magnetic susceptibility coefficient χ

- $\chi > 0$ (Paramagnetic materials):
Material enhances the magnetic field weakly and only while the external field is present.
→ Example: Olivine, Biotite.
- $\chi \gg 0$ (Ferromagnetic materials):
Material enhances the magnetic field strongly and can retain magnetization even after the external field is removed (permanent magnetization).
→ Example: Magnetite, Iron.
- $\chi < 0$ (Diamagnetic materials):
Material generates a magnetic response opposing the applied field, producing a weak repulsion.
→ Example: Quartz, Rock Salt.
- $\chi \approx 0$ (Practically Non-Magnetic materials):
Material shows negligible response to magnetic fields.
→ Example: Air.

$$M = \chi H$$

$$B = \mu_0(H + M)$$

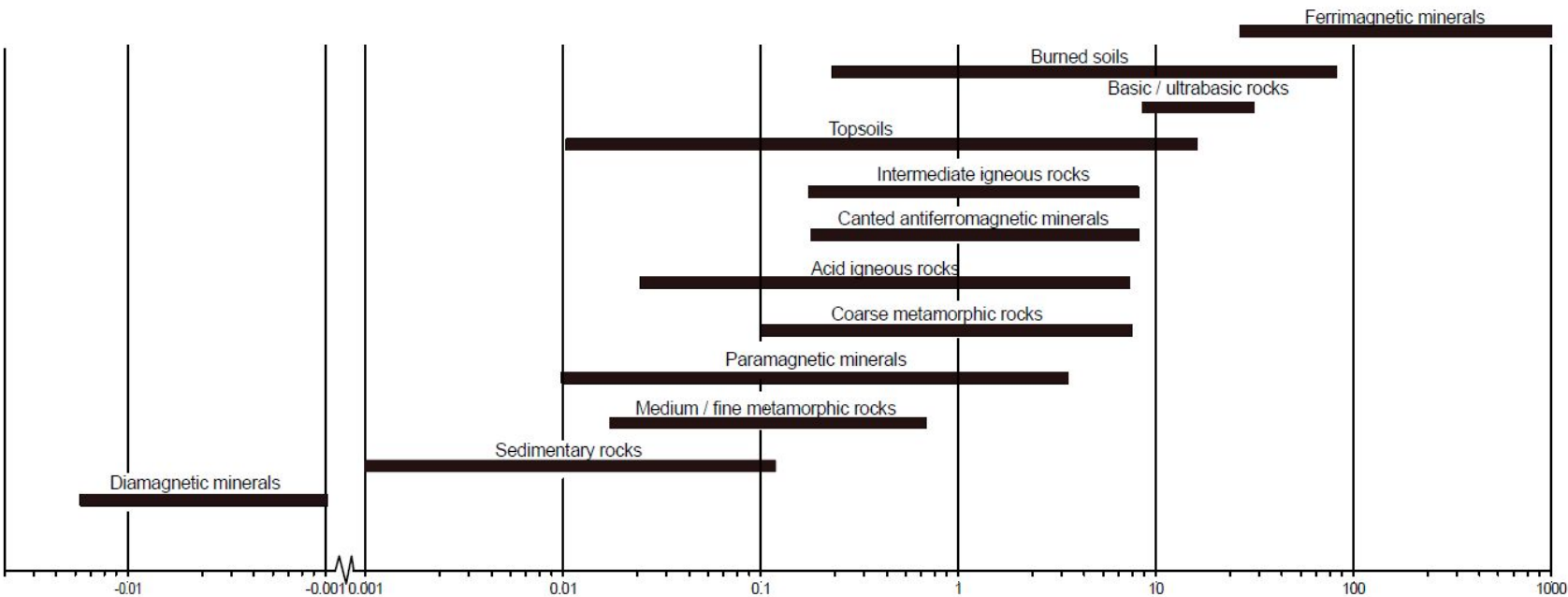
Material	Susceptibility x 10 ³ (SI)*
• Air	~0
• Quartz	-0.01
• Rock Salt	-0.01
• Calcite	-0.001 - 0.01
• Sphalerite	0.4
• Pyrite	0.05 – 5
• Hematite	0.5 – 35
• Ilmenite	300 – 3500
• Magnetite	1200 - 19,200
• Limestones	0 – 3
• Sandstones	0 – 20
• Shales	0.01 – 15
• Schist	0.3 – 3
• Gneiss	0.1 – 25
• Slate	0 – 35
• Granite	0 – 50
• Gabbro	1 – 90
• Basalt	0.2 – 175
• Peridotite	90 - 200

Magnetic Properties of Rocks

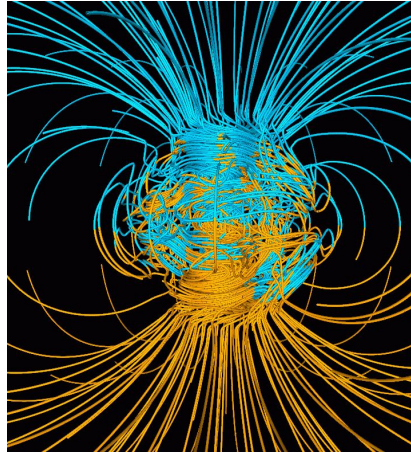
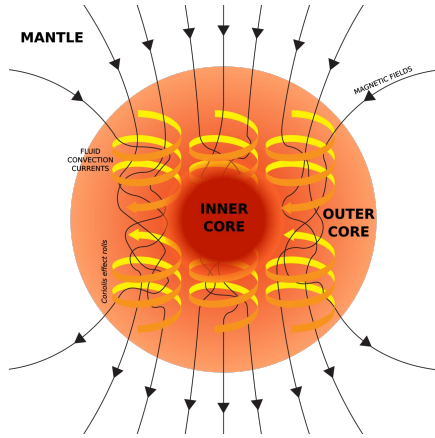
Igneous rocks often have high susceptibility (magnetite, iron-rich minerals).

Sedimentary rocks usually have low susceptibility.

Weathering reduces magnetic properties.



Earth's Magnetic Field Generation: The Geodynamo



Magnetic Field Simulation:

- Figure left : Convection rolls (yellow) and the generation of magnetic field lines.
- Figure right: Simulated magnetic field lines.
 - Blue lines: Field pointing inward.
 - Yellow lines: Field pointing outward.

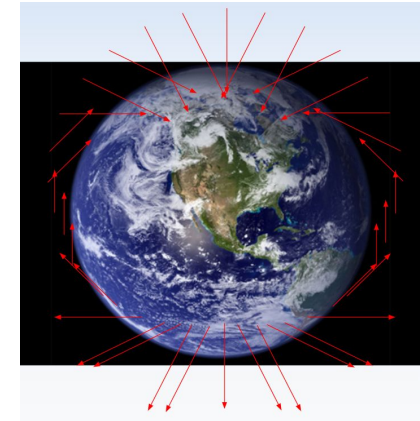
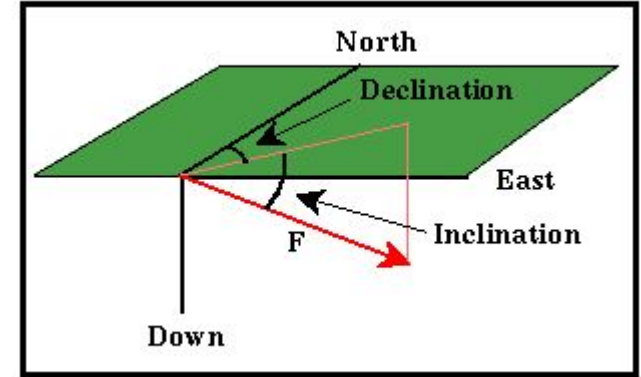
The Earth's magnetic field is generated by the motion of conductive fluids in the outer core.

- Convection currents of molten iron and nickel, driven by heat from the inner core.
- These flows are organized into rolls by the Coriolis force due to Earth's rotation.
- This moving fluid generates electric currents that produce magnetic field lines.
- The magnetic field resembles that of a dipole tilted with respect to the Earth's axis.
- The most complex field structure occurs in the liquid outer core.

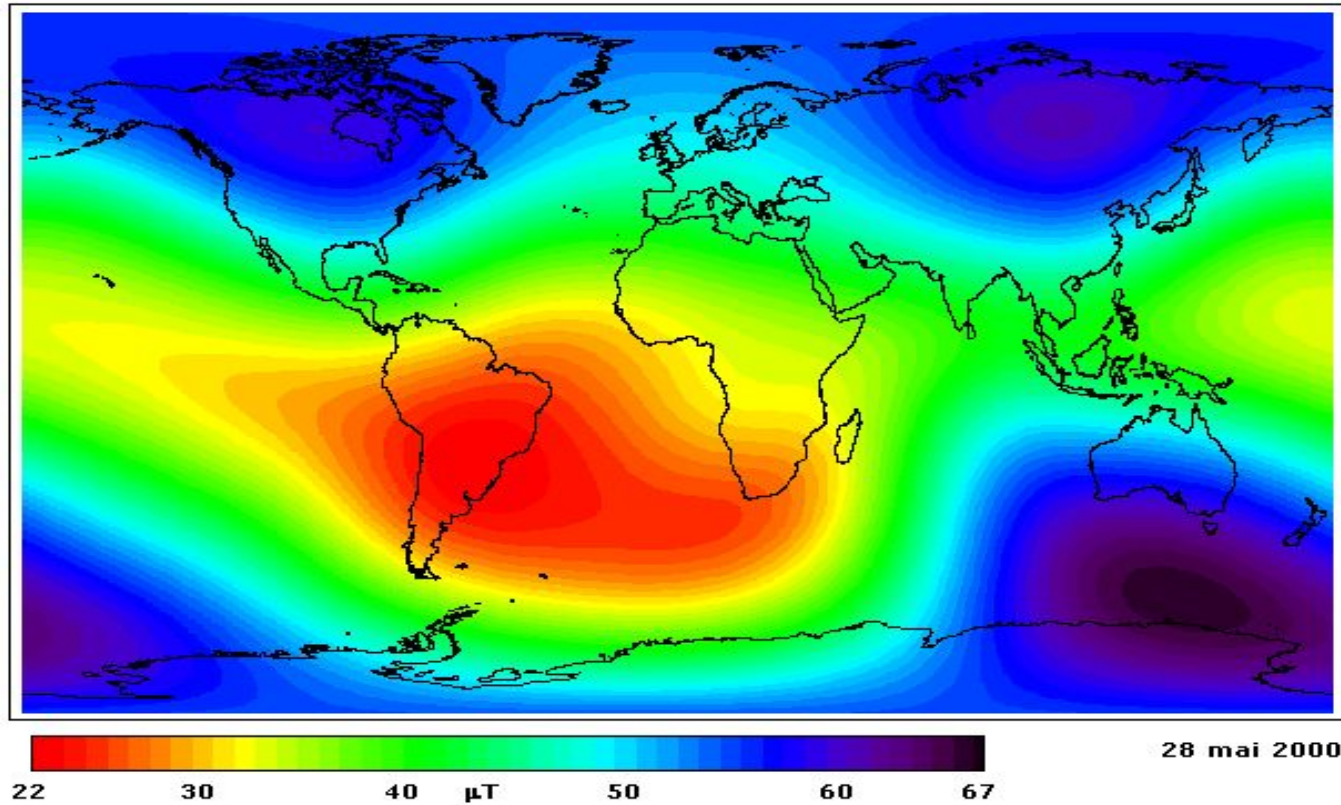
Magnetic Field: Intensity, Inclination, and Declination

At any point on the Earth's surface, the magnetic field, $\mathbf{F}$, has some strength and points in some direction. The following terms are used to describe the direction of the magnetic field:

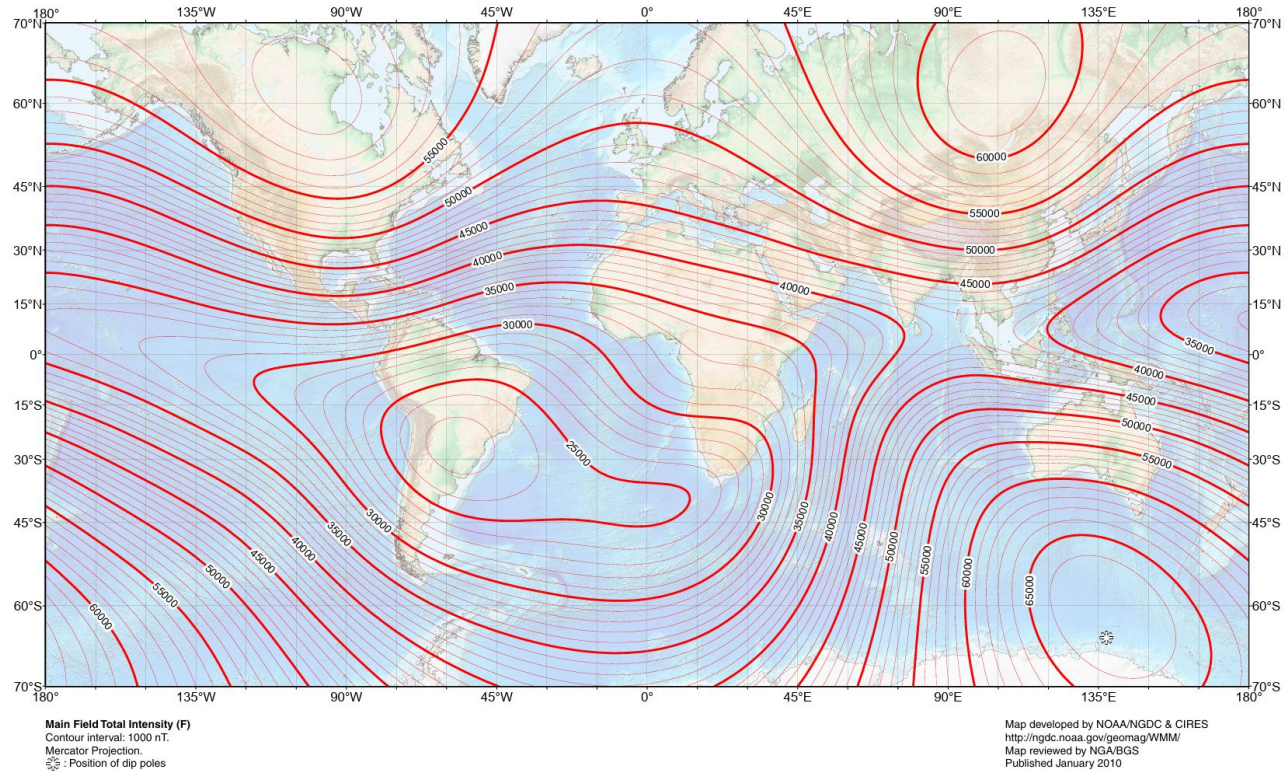
- **Intensity:** The strength of the magnetic field vector.
- **Inclination:** The angle between the magnetic field vector and the horizontal plane.
- **Declination:** The angle between geographic north and the horizontal projection of the magnetic field vector.
- About 90% of Earth's magnetic field behaves like a dipole aligned with the Earth's rotational axis.
- The remaining 10% cannot be explained by a simple dipole structure.
- The **International Geomagnetic Reference Field (IGRF)** is used to model the Earth's magnetic field.



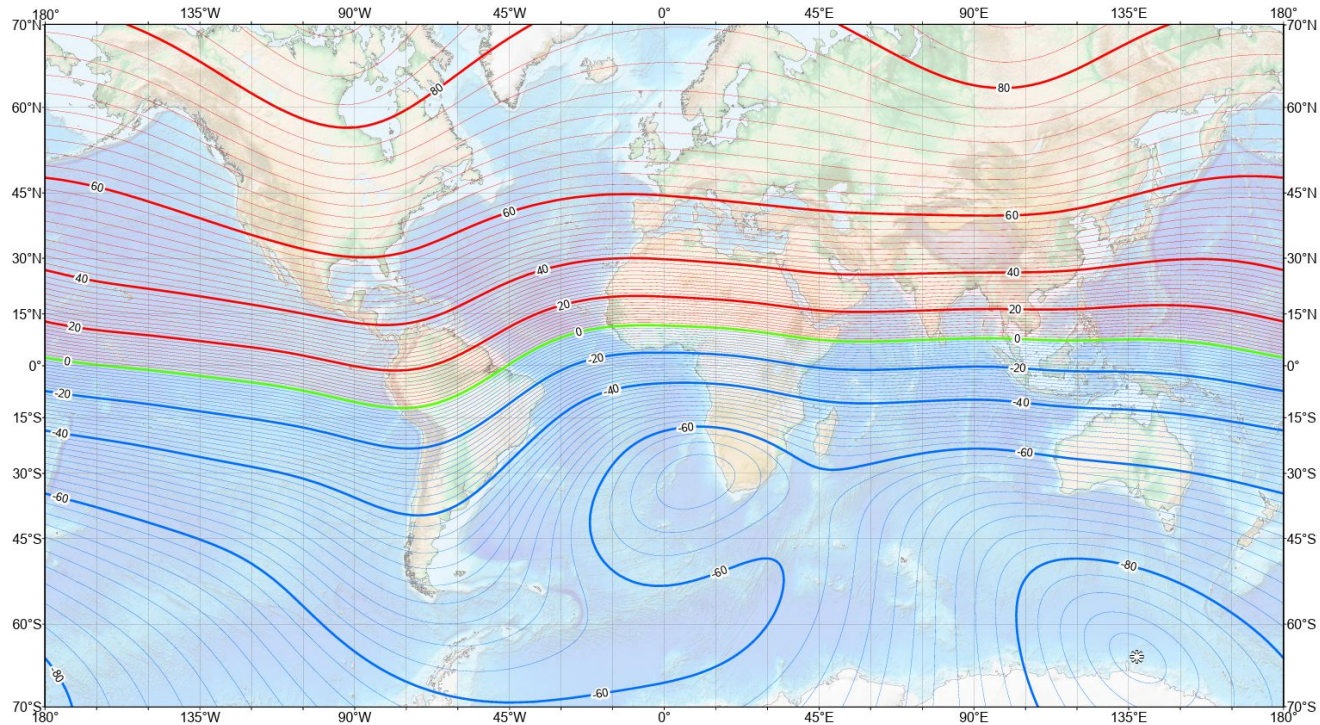
IGRF – Field Strength



US/UK World Magnetic Model -- Epoch 2010.0
Main Field Total Intensity (F)



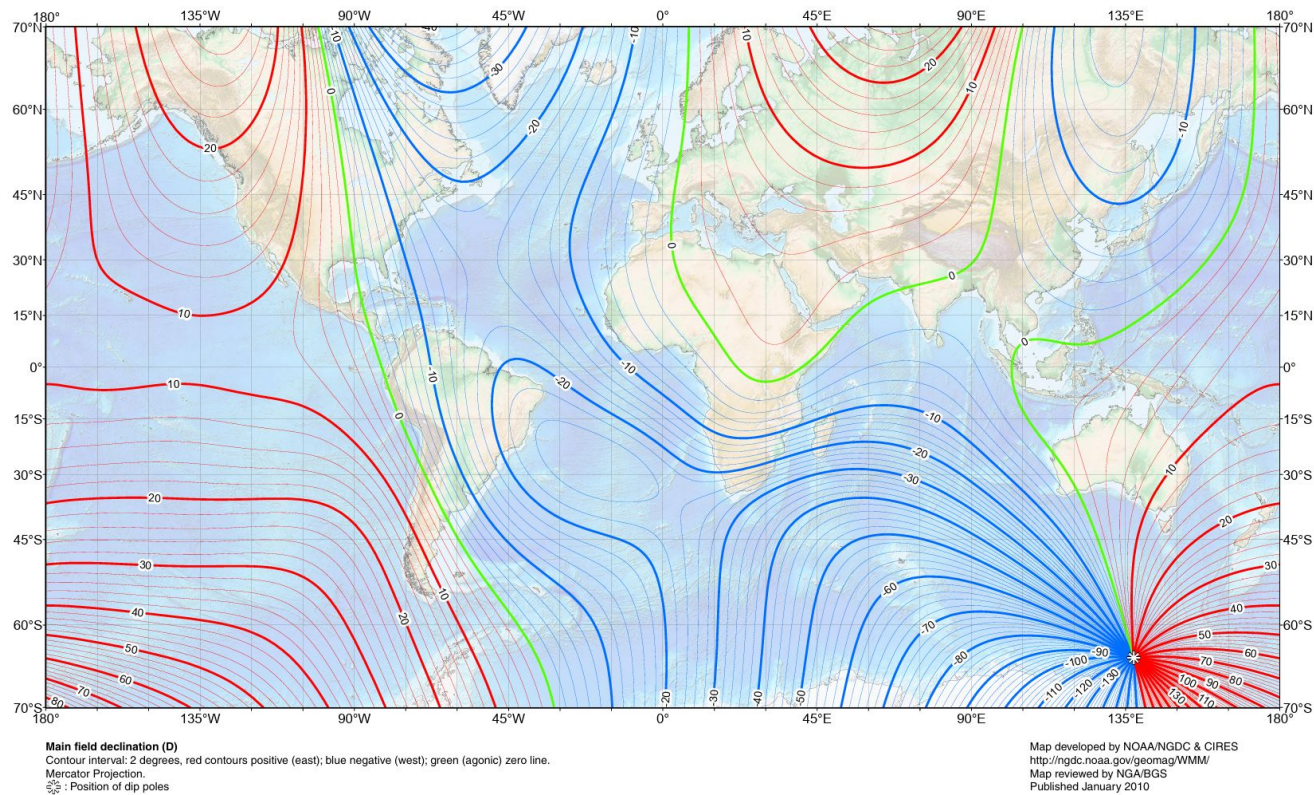
US/UK World Magnetic Model -- Epoch 2010.0 Main Field Inclination (I)



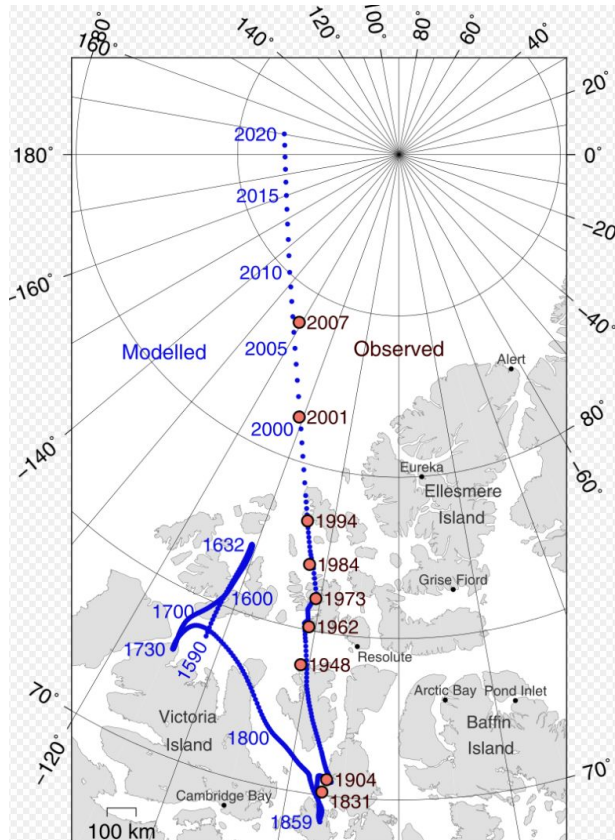
Main field inclination (I)
Contour interval: 2 degrees, red contours positive (down); blue negative (up); green zero line.
Mercator Projection.
☉☉ : Position of dip poles

Map developed by NOAA/NGDC & CIRES
<http://ngdc.noaa.gov/geomag/WMM/>
Map reviewed by NGA/BGS
Published January 2010

US/UK World Magnetic Model -- Epoch 2010.0 Main Field Declination (D)



Long-term secular variation: Movement of the Magnetic Pole



The Earth's magnetic pole moves over time (secular variation).

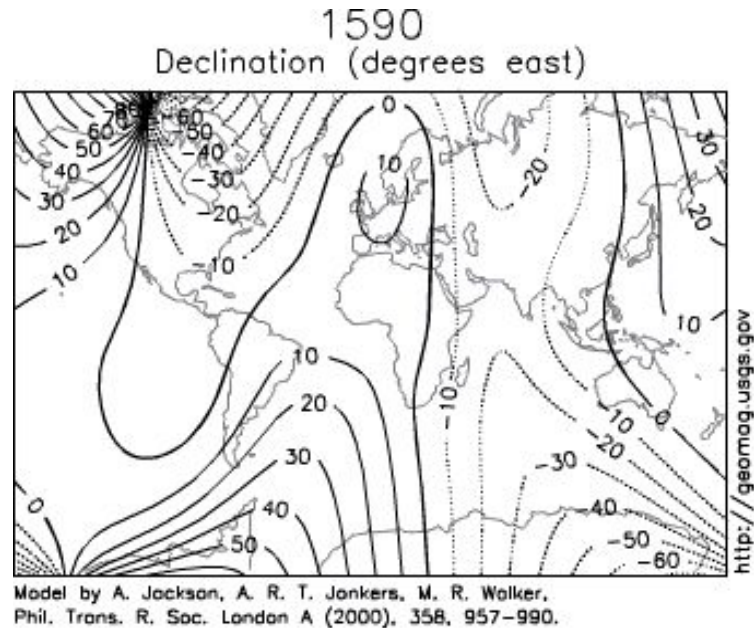
The red points show measured pole positions where the field is vertical.

The blue line shows modelled positions from magnetic field reconstructions.

In the last century, the pole shifted over 600 km towards the northwest.

Current movement is faster than 40 km per year.

Long-term secular variation: Changes in Magnetic Declination



The magnetic declination changes over centuries (secular variation).

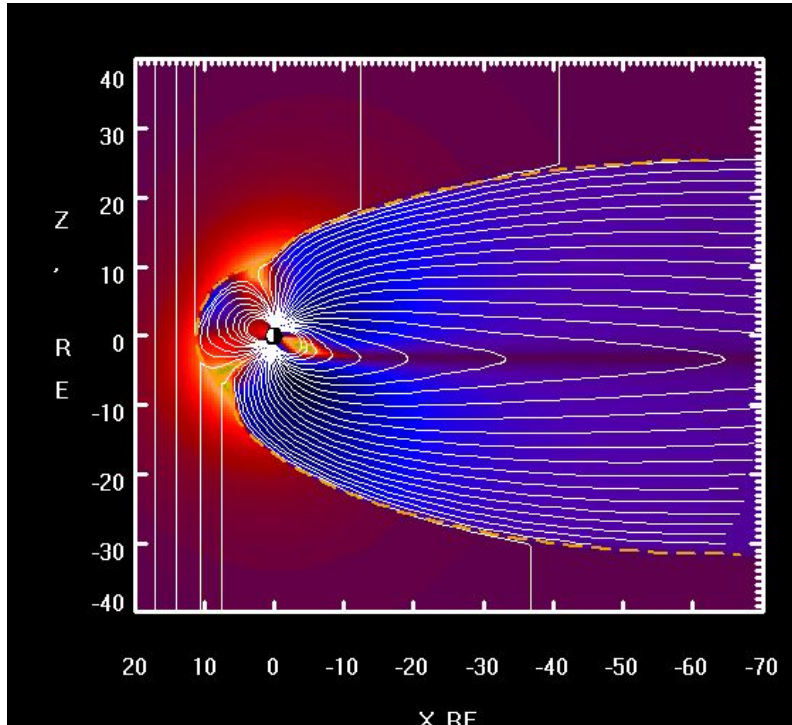
Declination is the angle between magnetic north and geographic north.

Isolines of declination have shifted significantly between 1590 and 1990.

Changes of tens of degrees have been observed globally.

This variation is key for navigation and magnetic surveys.

Short-term variations: Geomagnetic response to solar wind



The Earth's magnetic field interacts continuously with the solar wind.

The magnetosphere is compressed on the sun-facing (daytime) side.

The magnetosphere extends on the night side, forming the magnetic tail.

These changes produce short-term variations in the geomagnetic field (**diurnal variations**).

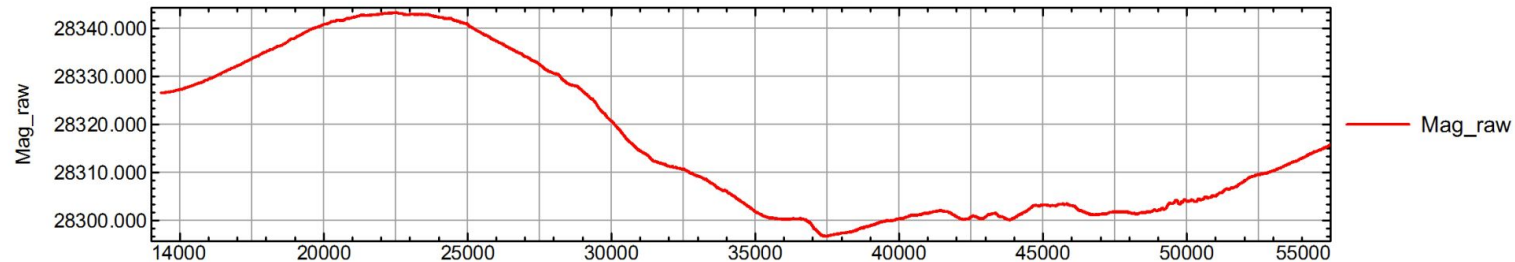
Short-term Magnetic Variations – Diurnal Cycle

The Earth's magnetic field changes daily due to solar radiation and ionospheric currents.

Typical diurnal variations range from 10 to 50 nT in quiet geomagnetic conditions.

The variation follows a smooth cycle that repeats each day.

These changes must be considered when interpreting magnetic survey data. → **Base station**



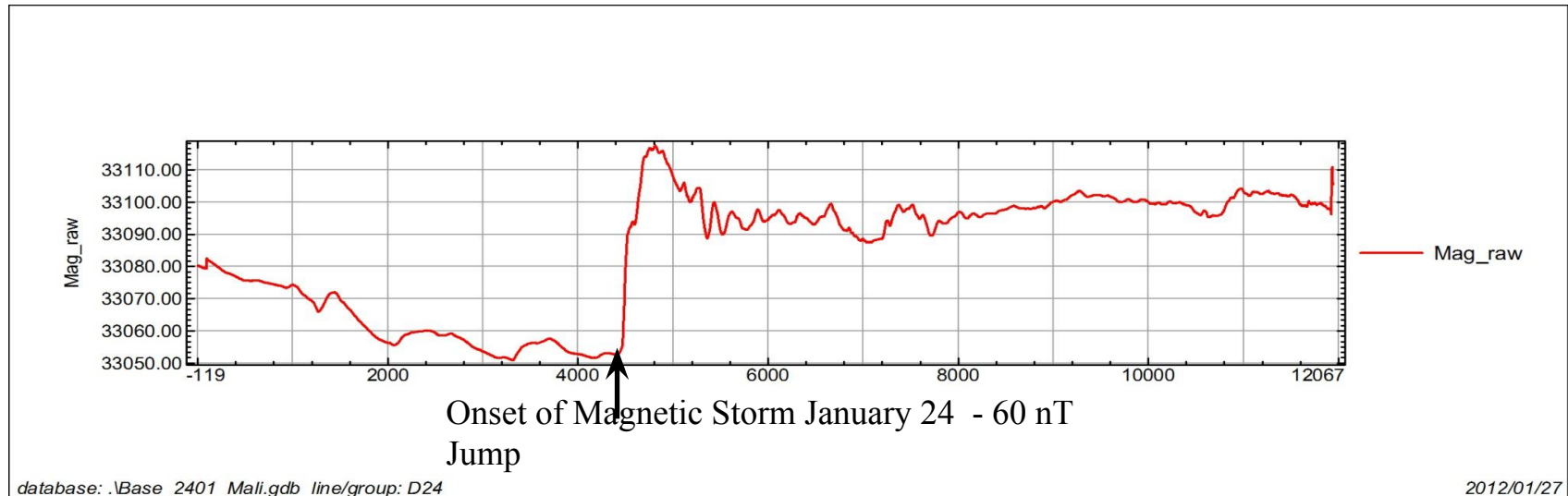
Short-term Magnetic Variations – Diurnal Cycle

Magnetic storms can cause **sudden increases** in the magnetic field.

Example: **~60 nT jump** in a few minutes during a magnetic storm onset.

These storms are caused by solar wind disturbances impacting the magnetosphere.

The magnetosphere compresses on the sun-facing side and distorts the Earth's field.



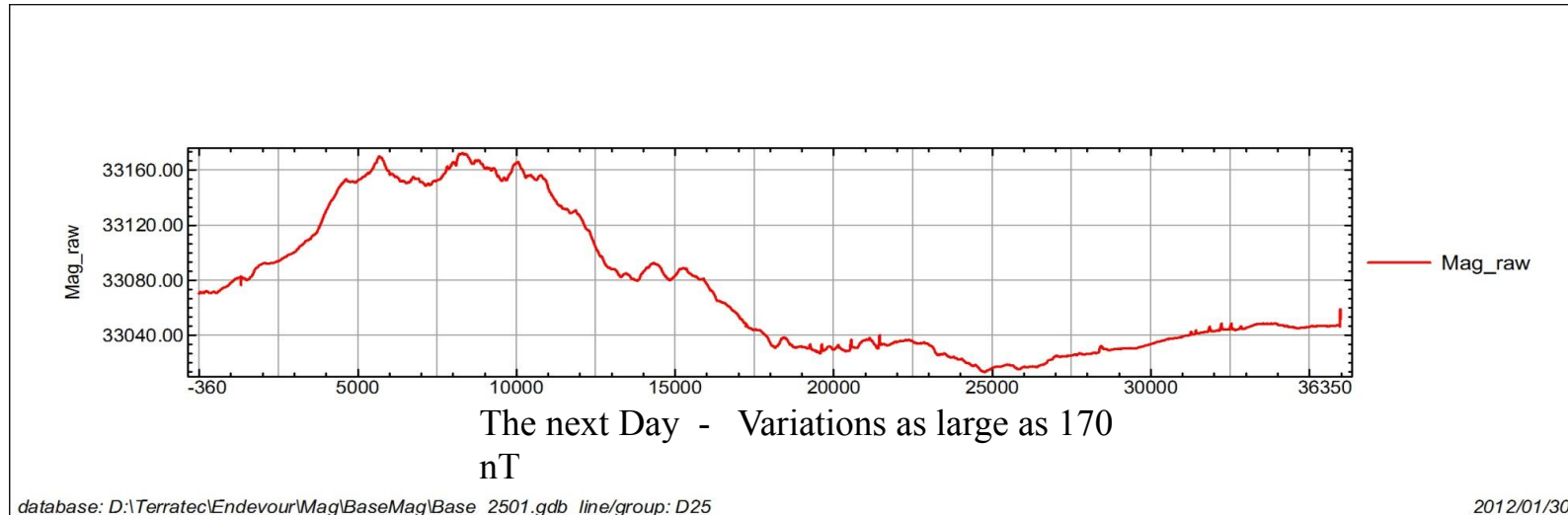
Short-term Magnetic Variations – Diurnal Cycle

After the onset, magnetic storms can **persist for hours or days**.

Example: Daily variations can reach **up to 170 nT** after a storm.

These intense variations can greatly exceed typical diurnal changes.

Magnetic storms significantly affect magnetic measurements and require careful interpretation.



On the surface of the earth the magnetic field can be broken into three separate components

Main Field

- Largest and most stable component.
- Generated by electrical currents in the Earth's outer core.
- Modeled by the IGRF. the International Geomagnetic Reference Field.
- Ranges from ~25,000 to 65,000 nT.
- Changes slowly (timescale of years).

External Magnetic Field

- Generated by sources external to the Earth.
- Changes quickly (timescale of minutes).
- Typically ranges from 10 to 50 nT.

Local Magnetic Field (Anomalous Field)

- Produced by near-surface sources like rocks, buried objects, minerals, or archaeological remains.
- Main target in geophysical prospecting.
- Typically ranges from ~10 to 10,000 nT.

Ground Magnetometer System

Highly sensitive scalar sensor used to measure the **total magnetic field intensity**

$$B_{\text{measured}} = |\vec{B}| = \sqrt{B_x^2 + B_y^2 + B_z^2} \quad B_{\text{measured}} = B_{\text{Earth}} + B_{\text{anomaly}} + B_{\text{diurnal}} + \varepsilon$$

- High-resolution measurement of the **geomagnetic field** (sensitivity: ~0.02 nT).
- Use as a **base station** to monitor diurnal variations (if stationary).
- Detection and mapping of **large-scale geological structures**, metalliferous mineral lodes, and archaeological remains.
- Locating **metallic objects**, buried pipelines, **military ordnance**, and toxic waste drums.



Ground Magnetometer System

Highly sensitive scalar sensor used to measure the total magnetic field intensity

$$B_{\text{measured}} = |\vec{B}| = \sqrt{B_x^2 + B_y^2 + B_z^2}$$

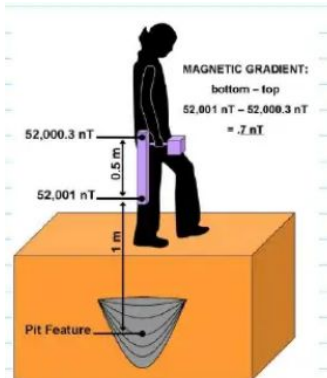
$$B_{\text{measured}} = B_{\text{Earth}} + B_{\text{anomaly}} + B_{\text{diurnal}} + \varepsilon$$

Vertical Magnetic Gradient →

$$\Delta B_z = \frac{B_{\text{top}} - B_{\text{bottom}}}{\Delta z}$$

- B_{top} : field measured by the upper sensor,
- B_{bottom} : field measured by the lower sensor,
- Δz : vertical distance between both sensors.

- Enhances the detection of shallow sources.
- improves the resolution of the anomaly shape and boundaries



Proton Precession Magnetometer

Phase 1: Alignment (Excitation phase)

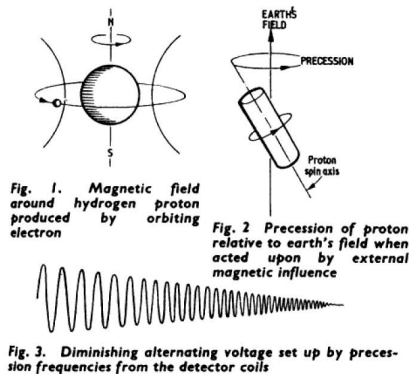
- A direct current (DC) is applied to the coil.
- This creates a strong magnetic field (B) perpendicular to the Earth's field.
- Protons inside the hydrogen-rich fluid align with this artificial field.

Phase 2: Precession (Measurement phase)

- The current is switched off.
- Protons try to realign with the Earth's magnetic field.
- As they realign, they precess (spin around the Earth's field direction, like a spinning top).
- This precession induces an alternating voltage in the detection coil.

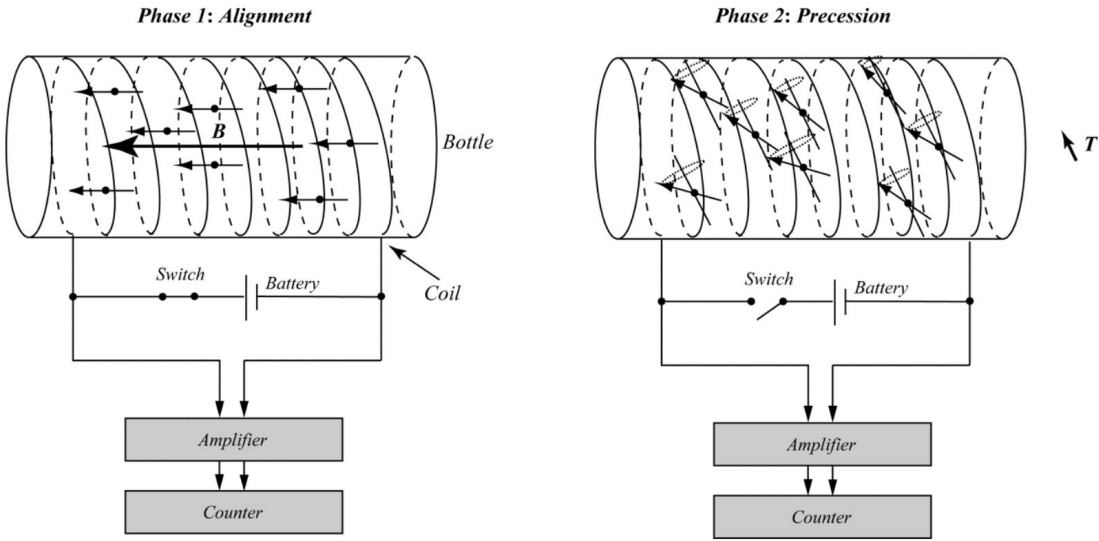
Finally:

- The precession frequency is directly related to the Earth's magnetic field strength (Larmor frequency).
- By measuring this frequency, the magnetic field intensity is calculated.



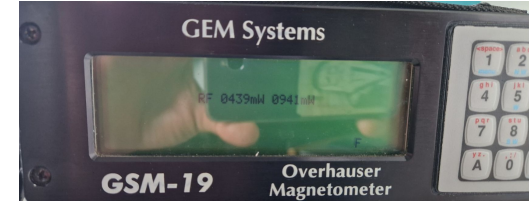
Magnetic properties of protons:

- Each proton has its own magnetic moment (like a tiny magnet).
- In the presence of a magnetic field, these magnetic moments tend to align with the field.



Overhauser Magnetometer: Improved Proton Precession

- Based on enhanced dynamic nuclear polarization
- Uses RF excitation instead of a strong DC magnetic field.
- More efficient excitation and much stronger signal.
- Consumes less energy, provides a continuous signal, and allows faster measurement times.
- Higher sensitivity: ~ 0.02 nT (vs. ~ 1 nT in proton systems)
- Higher sampling rate: Up to 10 Hz (vs. ~ 1 Hz in proton systems)



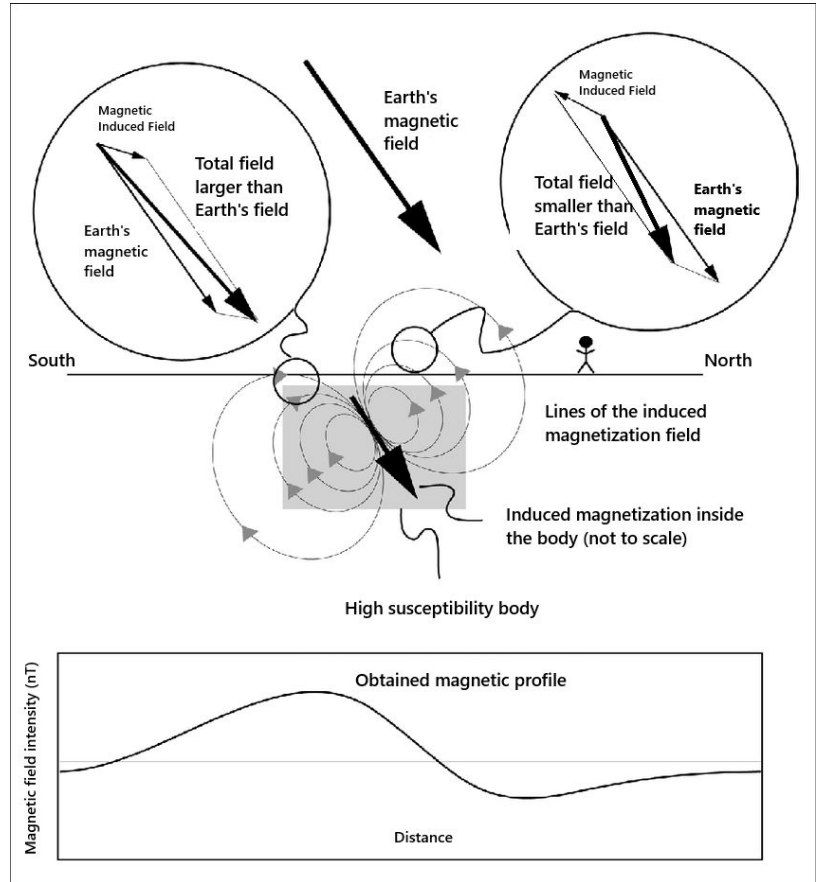
We encountered issues with the RF power level: it was lower than expected.

How Magnetic Anomalies are Generated

Magnetic anomalies are produced by bodies with higher magnetic susceptibility than the surrounding rocks.

The observed profile depends on:

- Magnetic susceptibility contrast between the body and the host rock.
- The body's shape, size, and depth.
- The Earth's magnetic latitude (north, equator, south).

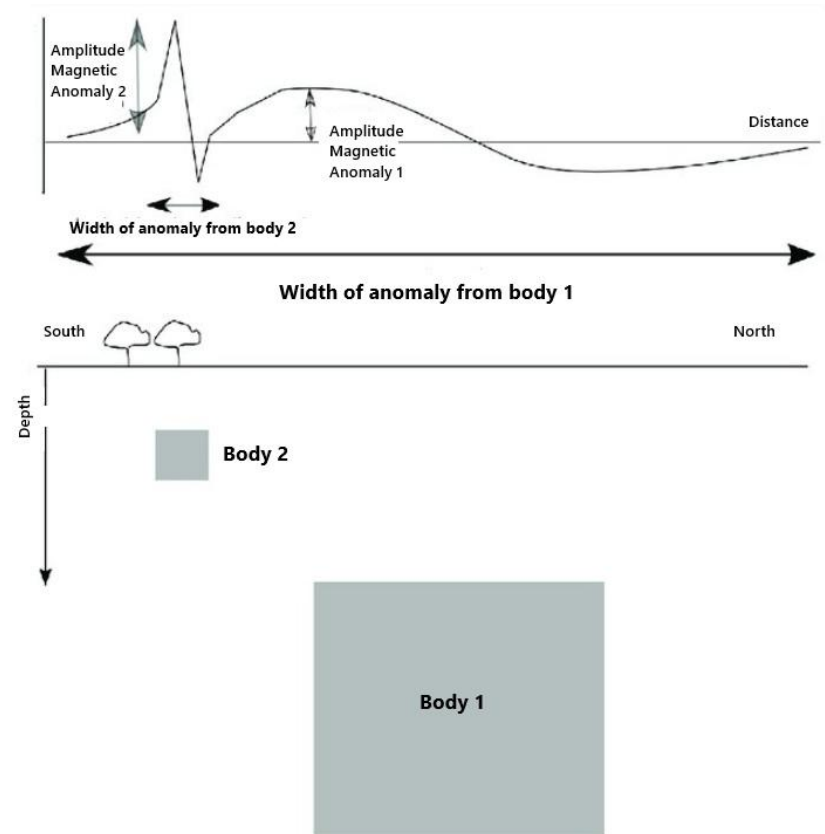


Effect of Body Size and Depth

Shallow, small bodies produce narrow, high-amplitude anomalies.

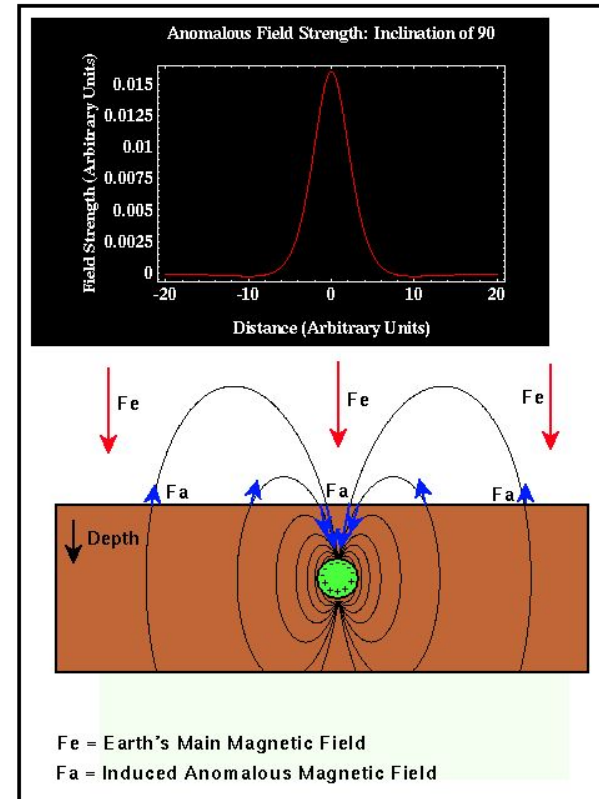
Deep, large bodies generate broad, low-amplitude anomalies.

The anomaly's wavelength gives clues about the depth.



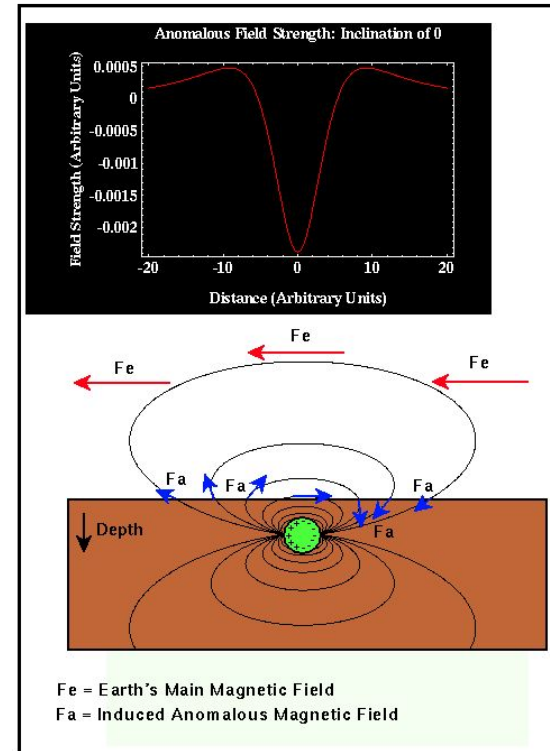
Magnetic Anomalies at the North Magnetic Pole

- At high latitudes, the Earth's magnetic field points straight down.
- The total field is the sum of the Earth's main field (F_e) and the induced anomaly (F_a).
- On the sides of the body, F_a opposes F_e , producing negative anomalies when F_e is removed.
- Above the body, F_a aligns with F_e , producing positive anomalies when F_e is removed.
- Induced anomalies show a central positive peak and negative flanks.

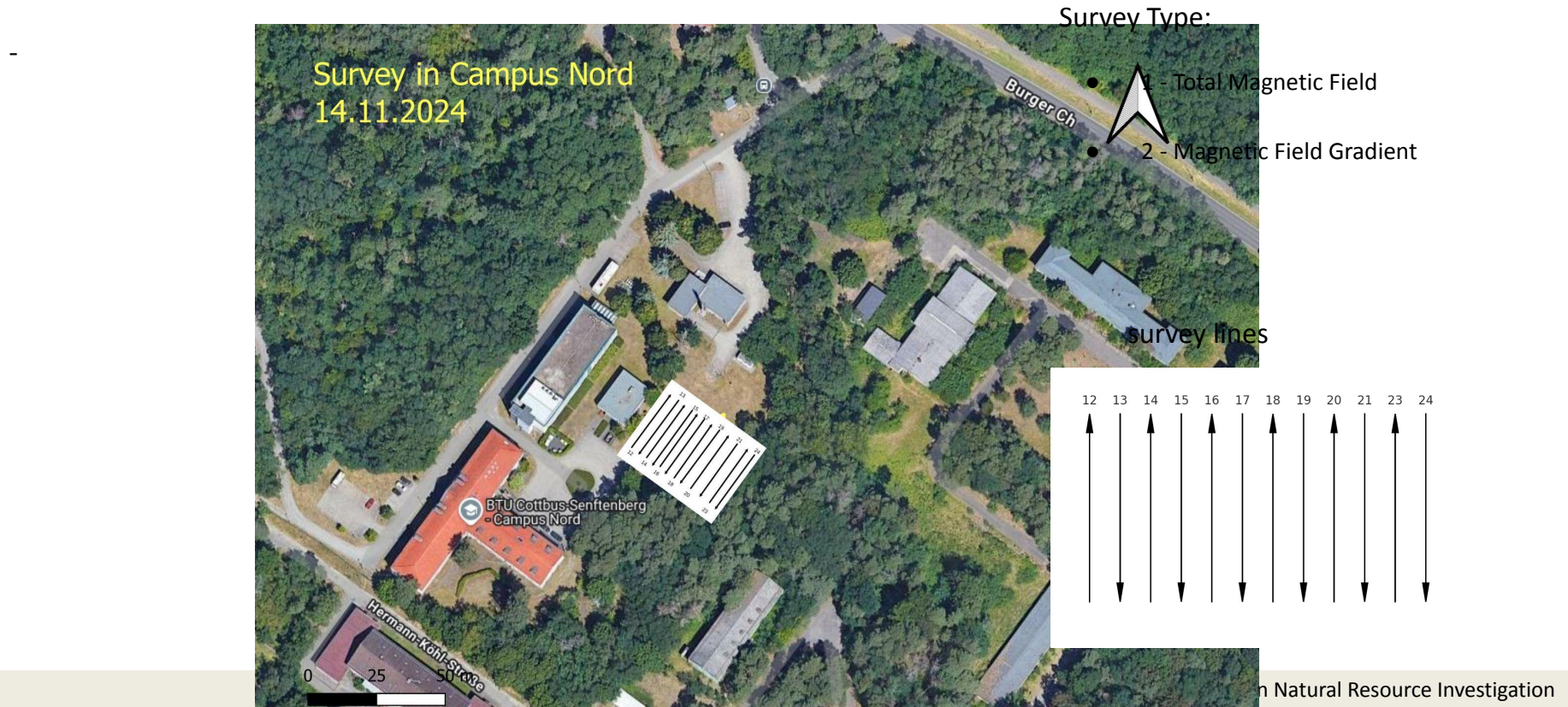


Magnetic Anomalies at the Magnetic Equator

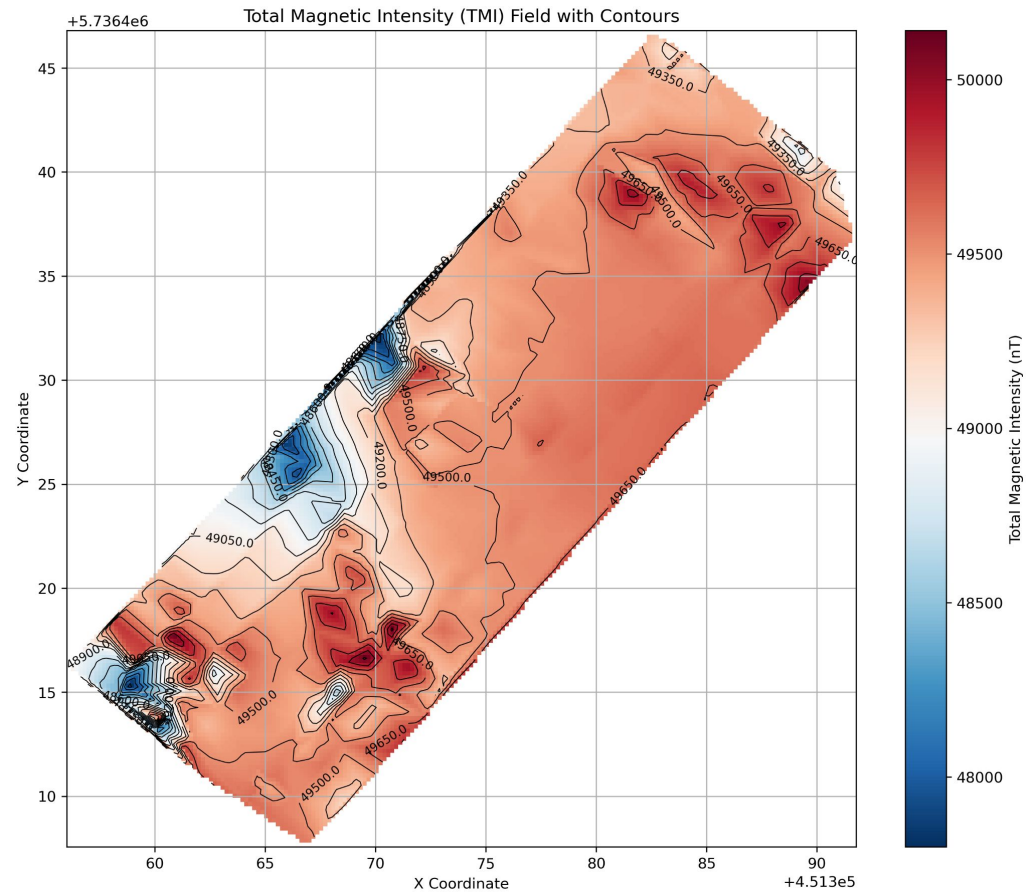
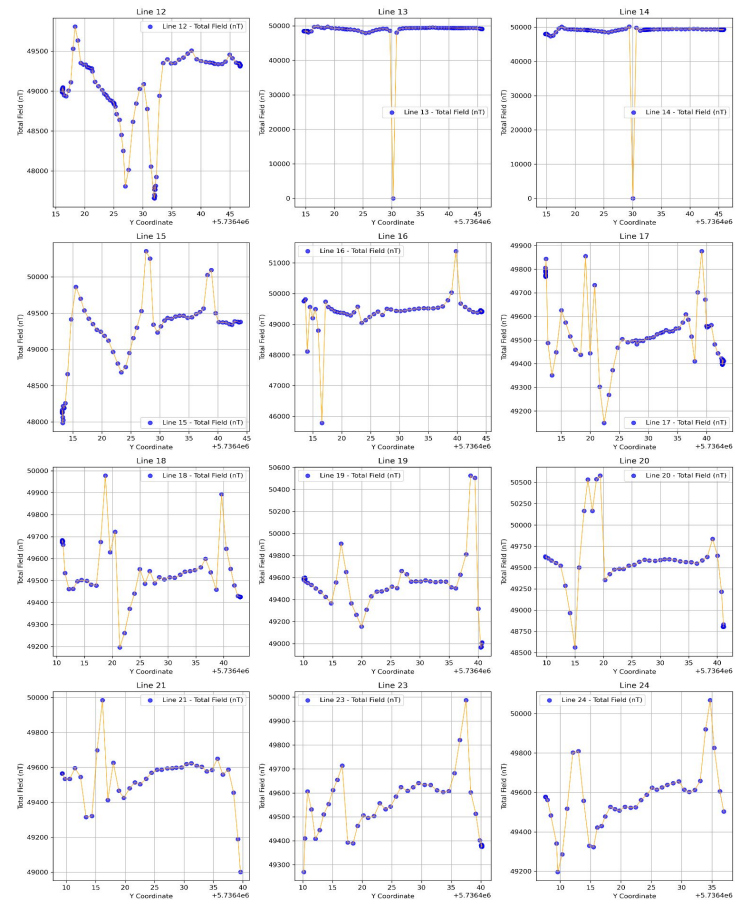
- The total field is the sum of the Earth's field (F_e) and the induced anomaly (F_a).
- On the sides of the sphere, F_a opposes F_e , creating negative anomalies.
- Above the sphere, F_a aligns with F_e , creating positive anomalies.
- When F_e is removed, we observe both positive and negative anomalies.
- The anomaly is symmetric: a central positive peak with smaller negative flanks.



Previous survey - Detection of local magnetic anomalies around campus structures



Total magnetic field results



Gradient magnetic field results

